### 1.7.4 Actors and the network

How do Akka actors communicate with each other across the network? `ActorRefs` are essentially addresses to actors, so all you need to change is how the addresses are linked to actors. If the toolkit takes care of the fact that an address can be local or remote, you can scale the solution by configuring how the addresses are resolved.

Akka provides discovery out of the box—which you'll learn about in chapters 6 and 9—that enables the transparency you seek. Akka passes messages for a remote actor to a remote machine where the actor resides and passes the results back across the network.

The only thing that has to change is how references to remote actors are looked up; this can be achieved solely through configuration, as you'll see later. The code stays the same, so you can often transition from scaling up to scaling out without having to change a single line of code. The flexibility of resolving an address is heavily used in Akka, as you'll learn throughout this book. Remote actors, clustering, and even the test toolkit use this flexibility.

At this point, you should understand that actors can give you more flexibility at a decent level of complexity, making your application far easier to scale. But there's a lot more to be learned, and as always, the devil is in the details. Let's get up and running with actors in the next chapter!

## Summary